

BIM 3D Object IFC item vis-à-vis QSBar item , and generating new product Particular Specification

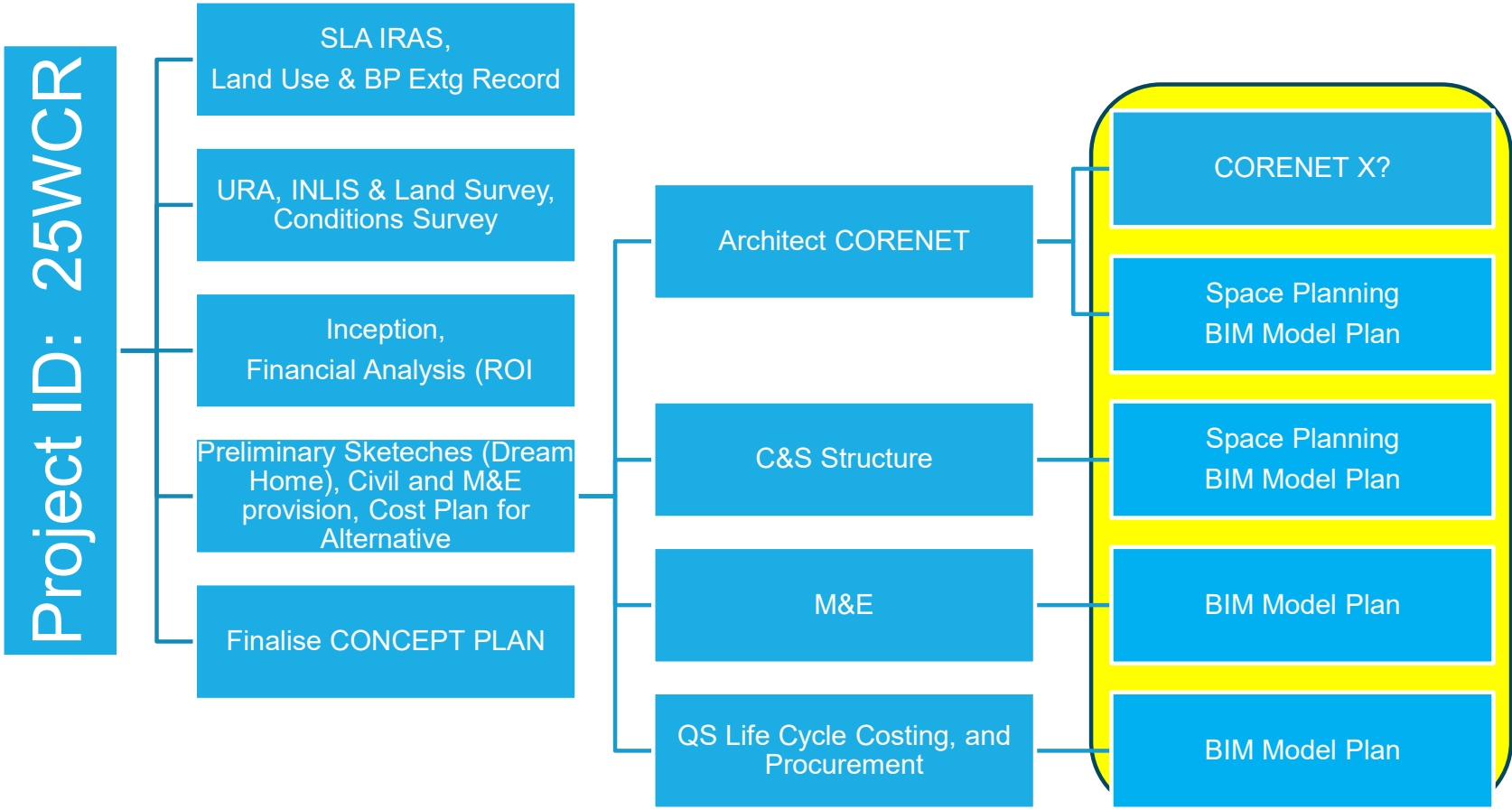
Team 2,
Team Lead: SEAH Kwee Yong
(BIM mgr, Cyril K H Seah Architects)

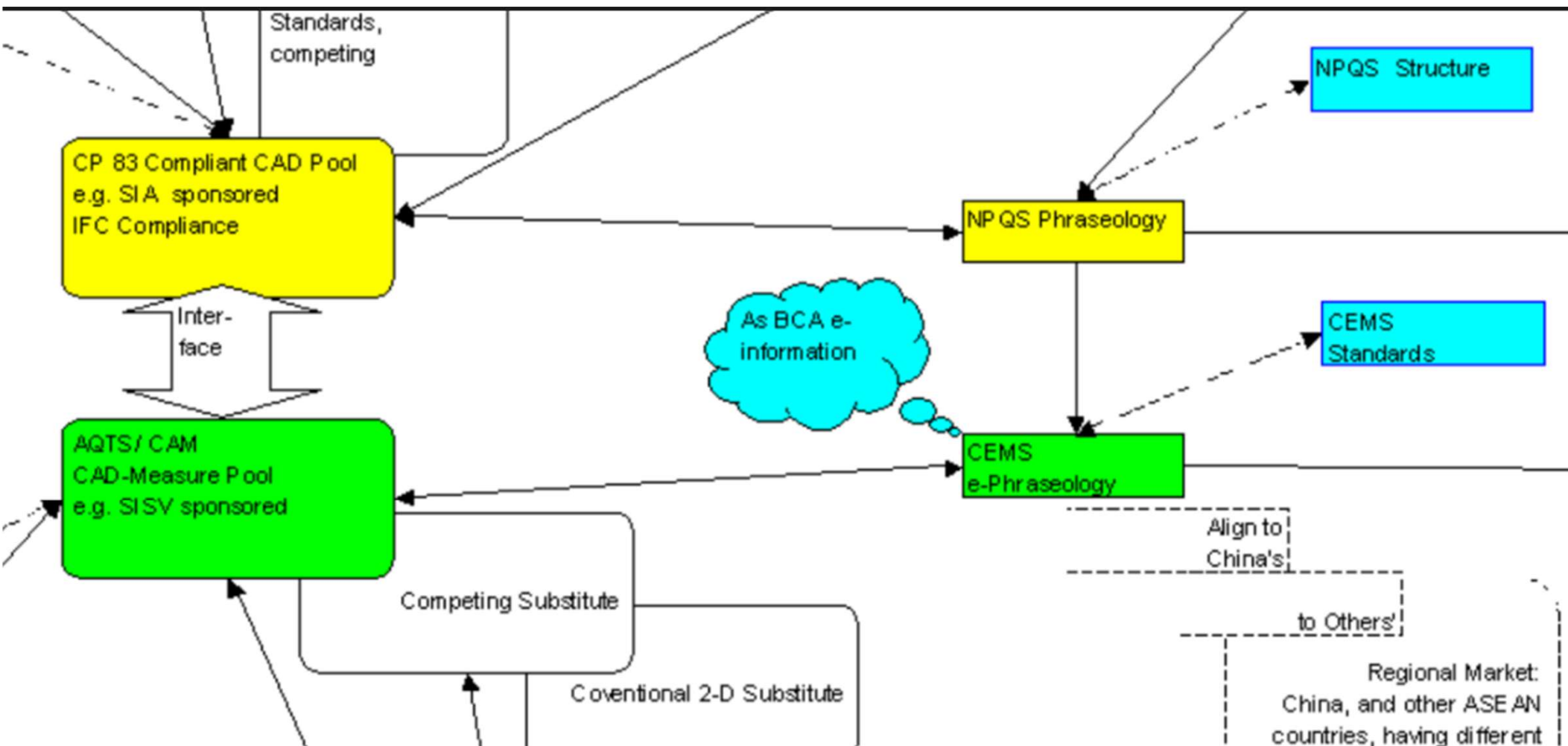
- **Members:**
- **CHENG Tai Fatt (CORENET Specialist),**
- **Gary ANG Choon Li (Business Analyst)**
- **Jacky CHIN Kong Hin (QS, ATS Advisory Services)**
- **LIM Wai Sing (Civil & Structural Engineer)**
- **Daniel CHOO (M&E Engineer)**
- **Bernard SEAH (Interior Designer)**

AI Hackathon 2025



Context of BIM





BIM 3D Object → 5D QSbar/Spec → 6D IFM Checklist

MVP: BIM 3D (Object) → 5D (QSbar/Spec) → 6D (Facilities Mgt)

BIM 3D Object IFC item vis-à-vis QSBar item, and generate phraseology for new product Particular Spec

ATS Advisory Services/

Cyril K H SEAH Architects

Google Trailblazer “Company 136” sandbox



Scope and Objectives



Approach & Tools



Strategies





Dataset

- **### *Step 1: Data Preparation***
 - - *Format Standardization*: - Convert documents into structured formats (e.g., *CSV, JSON, BIM/IFC, PDF with OCR*) for AI processing.
 - - Use *BCA's CORENET X* schema for BIM/GIS integration. - *Anonymization (if needed)*:
 - - Remove personally identifiable information (PII) from transaction data (e.g., *URA property transactions) to comply with **PDPA**.

Language of
BIM = IFC

to CSV format

Industry
Foundation
Classes for 3D
CAD
Interoperable
with downstream
software
(Smartbuilding)

industry-mapping-29-nov-2024 - View-only

	A	B	C	L	M	N	O
1	Agency	Identified Component	Identified parameter	Property Name	Property Type	Property Unit	IFC4 Material S
940	PUB	Water Meter	Length	Length	Length	mm	N.A
941	PUB	Water Meter	Purpose	Purpose	Label	N.A	N.A
942	PUB	Water Meter	Unit Number	UnitNumber	Label	N.A	N.A
943	PUB	Water Meter	Unit Number Tag	UnitNumberTag	Boolean	N.A	N.A
944	PUB	Water Meter	Water Supply Source	WaterSupplySource	Label	N.A	N.A
945	URA	Window	Operation Type	OperationType	Label	N.A	N.A
946	BCA	Window	Inner Diameter	InnerDiameter	Length	mm	N.A
947	BCA	Window	Outer Diameter	OuterDiameter	Length	mm	N.A
948	URA	Window	Material	Material	Label	N.A	Name
949	SCDF	Window	Safety Barrier Height	SafetyBarrierHeight	Real	N.A	N.A
950	SCDF	Window	Fire Access Opening	FireAccessOpening	Boolean	N.A	N.A
951	BCA	Window	Structural Height	StructuralHeight	Length	mm	N.A
952	BCA	Window	Structural Width	StructuralWidth	Length	mm	N.A
953	BCA	Window	Percentage of Opening	PercentageOfOpening	Real	N.A	N.A
954	BCA	Window	N.A	N.A	N.A	N.A	N.A
955	BCA	Window	N.A	N.A	N.A	N.A	N.A

Processes

- Start with Autodesk “BIM 3D Revit Export” to “IFC format” – Land survey, Architectural Revit, Structural Revit, M&E Revit
- Import into RIB BIM 5D CostX – To pick up the Revit exported IFC Model – For “Dimension Group” linkages;
- Bring in CorenetX QSBAR Phraseology – Convert from to Excel 4 layers structure, and then export as CSV format
- POC Test run. Found that Prompt upload can take CSV format (but crude).
- POC Context with Non-IFC all workable. Imagen can also be used

CSV for URA parameters

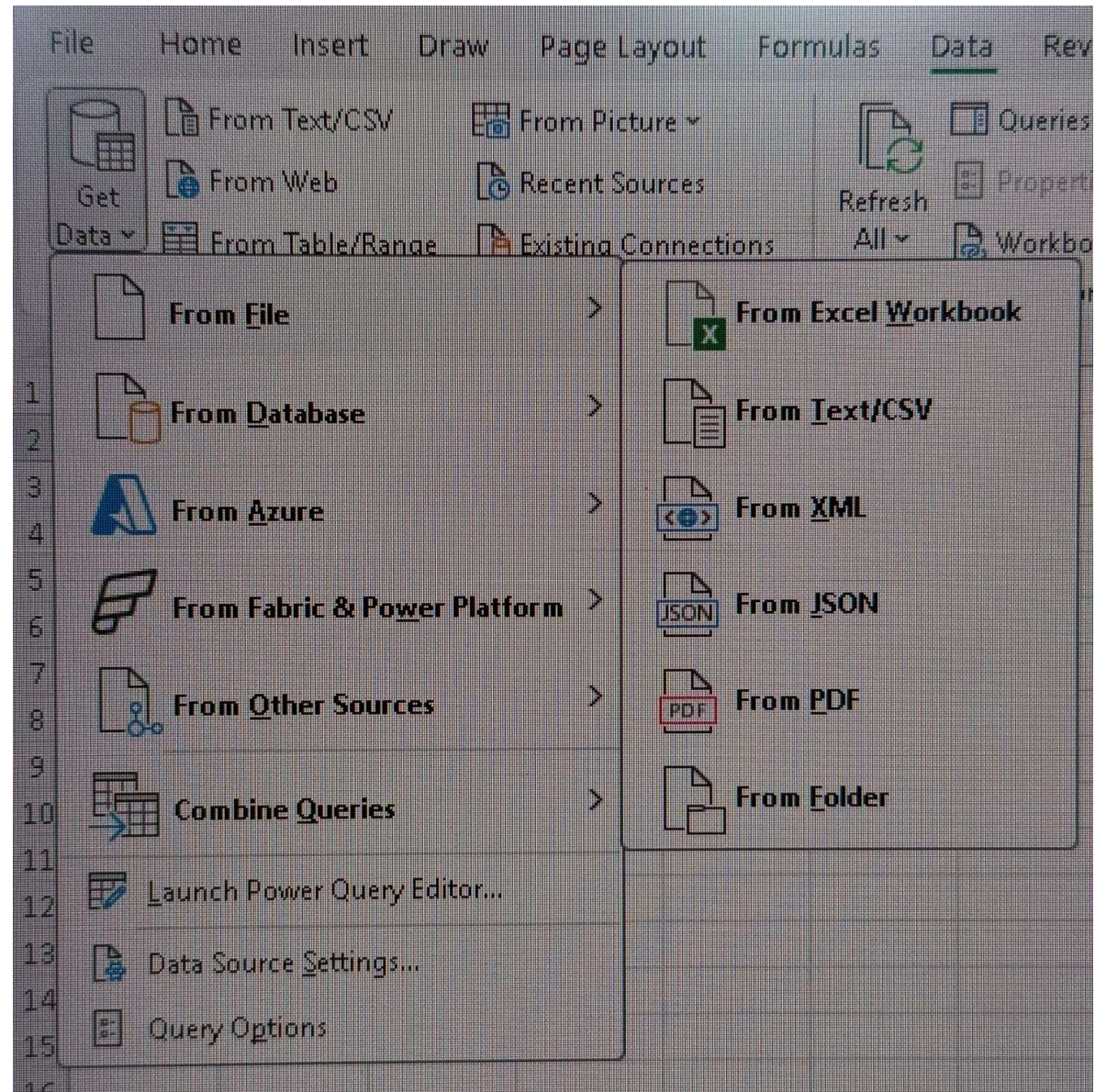
Overview of Non-URA spaces and sample value listing

Note: Sample values of the properties can be found in the next tab "Sample values for ref". Spaces may include values not indicated in this excel file.

ifcSpace sub-types	Need not specify	STRATALOT	COMMONPROPERTY	ACCESSORYLOT	Need not specify
Property Set	SGPset_Space	SGPset_StrataLot	SGPset_CommonProperty	SGPset_AccessoryLot	SGPset_SpaceDimension
Area_CommunityAndSportsFacilitiesS	AreaDescription	CommonArea	AccessoryLotArea	Area	
AutoIgnitionTemperature	BuildingName_Source	GeocodeID	AccessoryLotNumber	Height	
BarrierFreeAccessibility	Extg_ApprovedUse	TypicalFloor	AppurtenantTo	Volume	
BatteryType	Extg_StrataLotNumber		AreaDescription		
CategoryOfStorage	GeocodeID		BuildingName_Source		
ConstructionPhase	HouseNumber		GeocodeID		
ElderlyFriendly	IsVoid		HouseNumber		
EmergencyUse	LevelNumber		IsVoid		
ExternalReference	LotStatus		LevelNumber		
FloorFinishType	OldFormatLotNumber		LotStatus		
GasExtinguishingSystemType	ParcelType		OldFormatLotNumber		
HazardousMaterialClass	PlanNumber		ParcelType		
HearingEnhancement	PostalCode		PlanNumber		
Main	Prop_StrataLotNumber		PostalCode		
MaximumStoredEnergy	Prop_Use		StreetName_StreetCode		
NegativeRoomPressure	ResidingOnLandLot		SVYFileNumber		
Pool_TurnoverTime	StrataLotArea		TypicalFloor		
Pool_Volume	StrataLotNumber		UnitNumber		
Pool_WithRecyclingFacilities	StreetName_StreetCode		VoidArea		
PowerRating	SVYFileNumber				
PurposeGroup	TypicalFloor				
Retrofit	UnitName				
RoomPressure	UnitNumber				
SeatingCapacity	UnitType				
SpaceName	VoidArea				
StorageCategory					
SubmissionByFutureTenant					
ThresholdStoredEnergy					
TwentyFourHourMannedStation					

**Autodesk Revit can file
convert**

**(file format conversion
tools available)**



Use Case - Work around to upload IFC

(See our IFC to CSV demo Slides at the end of this Slides)

- So, we are building the app without going through the sandbox. Using our own Gemini API key.
- For the midpoint poc, we had written a **Streamlit script** to demo the following:
 - 1A. Upload IFC model
 - 1B. Use OpenAI and PandaAI to query filtered data.
 - 2A. Upload any pdf files (building codes, contracts etc)
 - 2B. Use **Gemini AI** to query and get response.
 - PandaAI - platform that transforms your natural language questions into actionable insights from your data sources. Connect your databases, use conversational AI to explore your data, and collaborate with your team-mates

Show specific columns – save to output.csv

```
- Classification
- Level
- Qto_WallBaseQuantities.id
- Qto_WallBaseQuantities.GrossSideArea
- Qto_WallBaseQuantities.GrossFootprintArea
- Qto_WallBaseQuantities.NetVolume
- Qto_WallBaseQuantities.NetSideArea
- Qto_WallBaseQuantities.Height
- Qto_WallBaseQuantities.Width
- Qto_WallBaseQuantities.Length
- Qto_WallBaseQuantities.GrossVolume
```

	A	B	C	D	E
1	Level	Qto_WallB	Qto_WallBaseQuantities.Width		
2	Level 1	2700	202		
3	Level 1	2700	202		
4	Level 1	2700	202		
5	Level 1	2700	165.1		
6	Level 1	2700	165.1		
7	Level 1	3900	202		
8	Level 1				
9	Level 1	3500	280		
10	Level 1	3500	280		
11	Level 1	3500	300		
12	Level 1	1500	280		
13	Level 1	1500	280		
14	Level 1	1500	280		
15	Level 1	3250	202		
16	Level 1				
17	Level 1	2700	165.1		
18	Level 1	3300	280		

```
-----
Enter your query about the CSV data (e.g., 'What is the largest GrossSideArea?', 'Show walls on Level 1'):
> Show all rows with column headers "Level", "Height" and "Width"
```

MPV: Limitation of Gemini 2.0 Flash

- Use of IFC --> CSV --> Excel --> Business Intelligence (Quite crude in AI)
- Gemini seems workable with non-IFC files. So, we could communicate with other non-IFC files of the project:
 - Generative Construction Spec Phraseology
 - **National Productivity Quality Specifications (NPQS)** – As foundation model
The Intelligent National Productivity and Quality Specifications (iNPQS) - Singapore Institute of Architects
 - Project Specific Document (PSD) Spec – as project DataSet
 - Generative Construction **Schedule of Rates (SOR)**
 - BCA Unit Rates on Web. With periodical data – As foundation model
https://eservices.bca.gov.sg/infonet/download/unitrates_sample.pdf
 - Project Specific Breakdown of Cost or priced SOR – as project DataSet
- Generate Response for new project/ product
 - Upload new construction material dataset
 - Prompt for **new “Spec paragraph” for the new product.**

File Uploaded as model foundation



ai-sb-user-811-gpu

File Edit View Run Kernel Git Tabs Settings Help



Filter files by name



/ trailblazer2 / FSOR /

Name	Last Modified
1. Builder's Works.pdf	a day ago
Part 1 Building Works - 347 pages.pdf	a day ago



Table of Contents



Launcher



Part 1 Building Works - 347 p x



ITEM NO.	DESCRIPTION	UNIT	RATES (\$\$)
D300000	BLOCKWORK Precast concrete vent block of any size complying with SS 76M, bedding, jointing and pointing in cement and sand (1:4) mortar reinforced with and including brick reinforcement at every courses		
D300008	100mm thick walls	m2	100.90
D300009	150mm thick walls	m2	116.70
D300010	100mm thick concrete vent block in grille blockwall consisting of rain-proof concrete vent blocks (any size) bedded and jointed in	m2	72.40

/ trailblazer2 / NPQS /

Name	Last Modified
Untitled Folder	a day ago
A01-010 BRICKWORK-FPC.pdf	a day ago
A01-020 BLOCKWORK-FPC.pdf	a day ago
A01-030 LIGHTWEIGHT CONCRE...	a day ago
A01-040 PRECISION BLOCKS-FPC....	a day ago
A01-050 PRECAST VENT BLOCKS-...	a day ago

2. PERFORMANCE REQUIREMENTS

2.1 Contractor's Brief

When carrying out the proposals as set out in Clause 1.4.1, take a following requirements:

2.1.1 Structural Criteria

2.1.1.1 Structural Movement and Movement Between Other Works

Test & Measures



Outcomes



Test: Generating a Particular Specification for new product (object) guided by NPQS uploaded

System instructions   

You are a design assistant bot for drafting Particular Specification from NPQS. Your job is to assess the needs of a first-time user so they can gain access to the NPQS and design a Particular Specification for a new product.

USER Hi! Id like to access the NPQS

AI Hello! I'm happy to assist you in accessing the NPQS (National Product Qualification System) and drafting a Particular Specification for your new product. To ensure we tailor the process to your specific needs, let's start with a few questions. This will help me understand your project and guide you through the necessary steps.

First, could you please tell me a bit about the product you're developing?



Refine response

Then, a product Catalogue can be uploaded for AI Response

AI Thank you for providing the document! Based on the information you've given me, you're looking to develop a Particular Specification for Dry Wall Partitions using the NPQS A02-010 document as a template.

To start drafting your specification, let's gather some key details about your specific product and project requirements. This will allow me to tailor the specification to your exact needs. Please answer the following questions as thoroughly as possible:

1. Project Scope:

- What type of building is this dry wall partition for (e.g., residential, commercial, industrial)?
- What is the overall size and complexity of the project (e.g., single room, entire floor, multiple buildings)?

"Youtube link" can be included in Prompt

USER Sample of my drywall partition



AI Thank you for providing that detailed video! It's very helpful to visualize the type of dry wall partition you are using. I would recommend we get you signed up for the standard package. This includes a more thorough consultation to ensure the specification meets your precise needs. This is what we need so I can give you the best and most custom fit.

Based on your video, it seems like we can outline the following:

Project Scope:

Our original objective is to upload the IFC (3D Object) turned CSV, to be recognised by Vertex AI Gemini as if it is a Youtube file attachment.

Another Test:

**Dataset of existing specification for Integrated Facilities Mgt (IFM).
Then, Generate for new IFM phrases**

- Train model
 - Existing Hawker Centre IFM spec
 - Existing National Heritage IFM spec
 - Existing MSE Steven Rd IFM spec
- Prompt to generate next project
 - Singapore Poly Campus' new IFM spec
- Response
 - Element by Element checklist (See Word Example)
 - Checklist for each Element listed

Generate **Element by Element**

Glossary of Terms

Chapter 1 What are the components for automatic lightning risk alert system (ALRAS) .	2
Chapter 2 What are the components of Building Automation System	4
Chapter 3 What are the components of Fire Protection System (FPS), Emergency Voice Communication System (EVCS) and Public Address (PA) System	5
Chapter 4 What are the components of Audio-Visual System (AV)	7
Chapter 5 What are the components of Vehicle Barrier System (VBS)	8
Chapter 6 What are the components of Card Access System (CAS).....	10
Chapter 7 What are the components of Closed Circuit Television (CCTV) Monitoring	

Chapter 1 What are the components for automatic lightning risk alert system (ALRAS)

The Automatic Lightning Risk Alert System (ALRAS) is designed to provide accurate and timely warnings about lightning risks. Here are the key components of ALRAS:

1. Detection Sensors

- **Electrostatic Field (E-Field) Sensors:** These sensors monitor the formation and dissipation of lightning by measuring the electrostatic field strength.
- **Electromagnetic (EM) Wave Sensors:** These sensors detect electromagnetic waves generated by lightning strikes, suitable for monitoring nearby or approaching lightning.

2. Central Processing Unit (CPU)

- **Data Analysis:** The CPU processes data from the sensors to determine the lightning risk based on electromagnetic field strength and distance of the discharge.

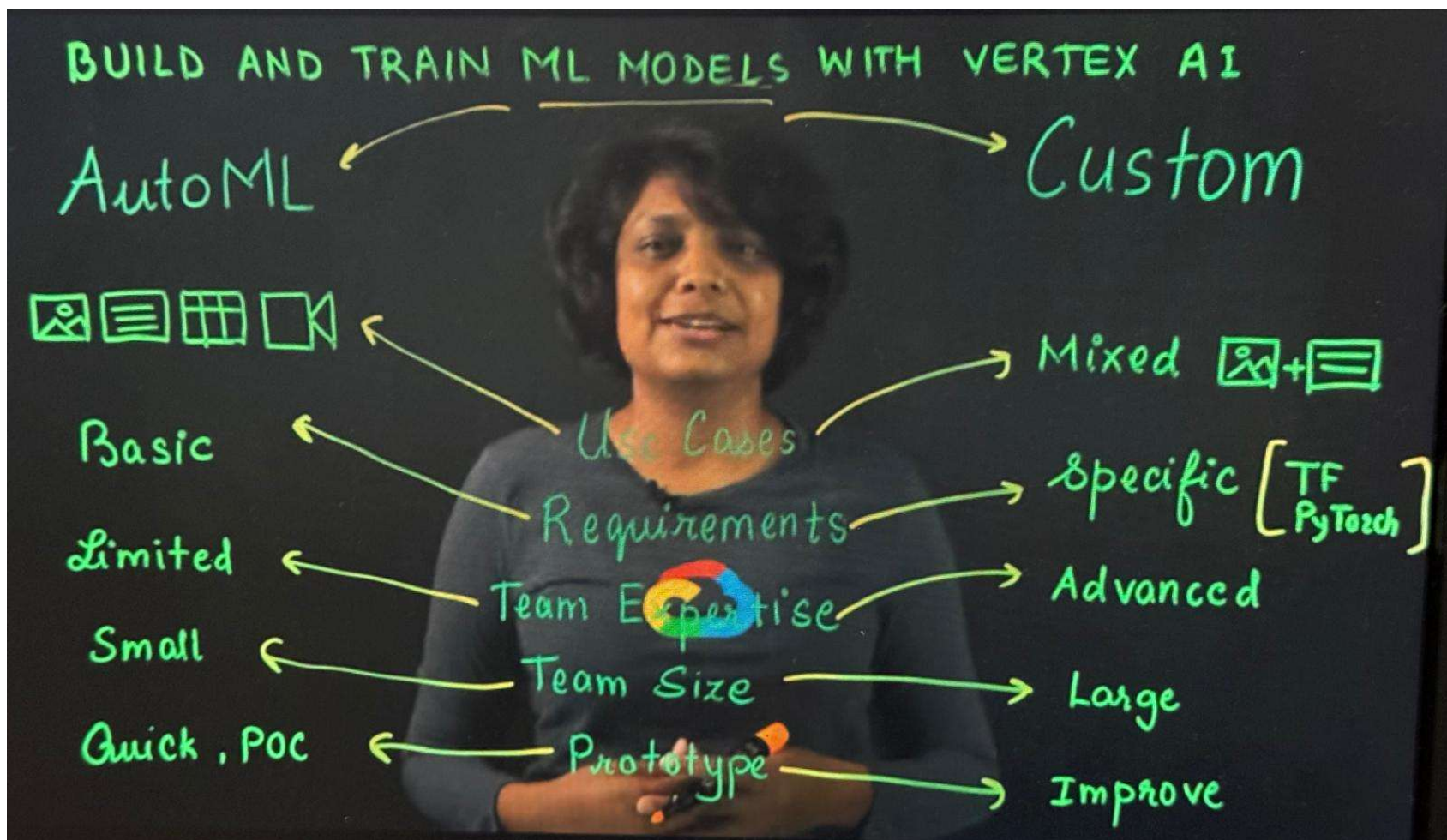
What other CAD vendors are doing?

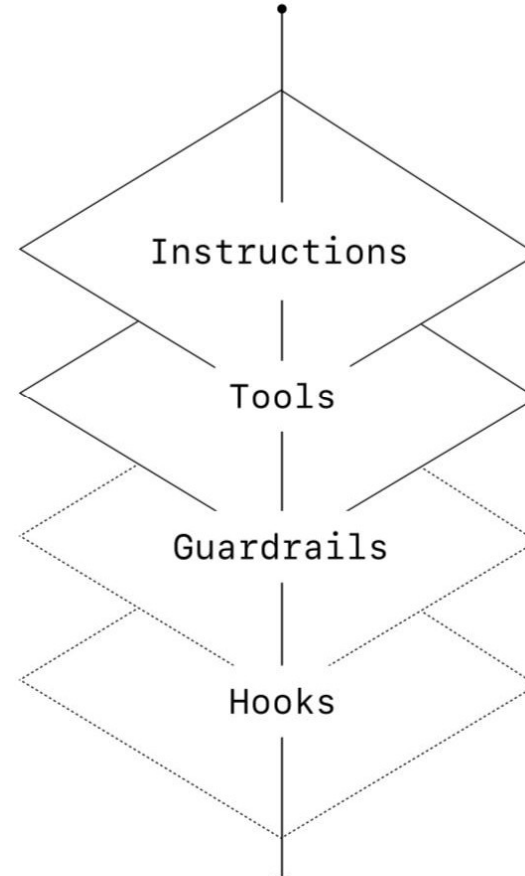
Bentley BIM 3D +Vertex AI Studio and Agent Builder

The synergy between Bentley and Google is further amplified by Google's Vertex AI, a fully managed AI platform powered by Google's Gemini models. Vertex AI offers developers tools like Vertex AI Studio and Agent Builder to deploy advanced capabilities at scale.

"Through our partnership, we are leveraging Vertex AI to innovate faster," Pittman says. "Google's infrastructure significantly enhances our analytical capabilities, enabling us to deliver swift and accurate insights that city managers can act upon immediately."

Train and Fine tune





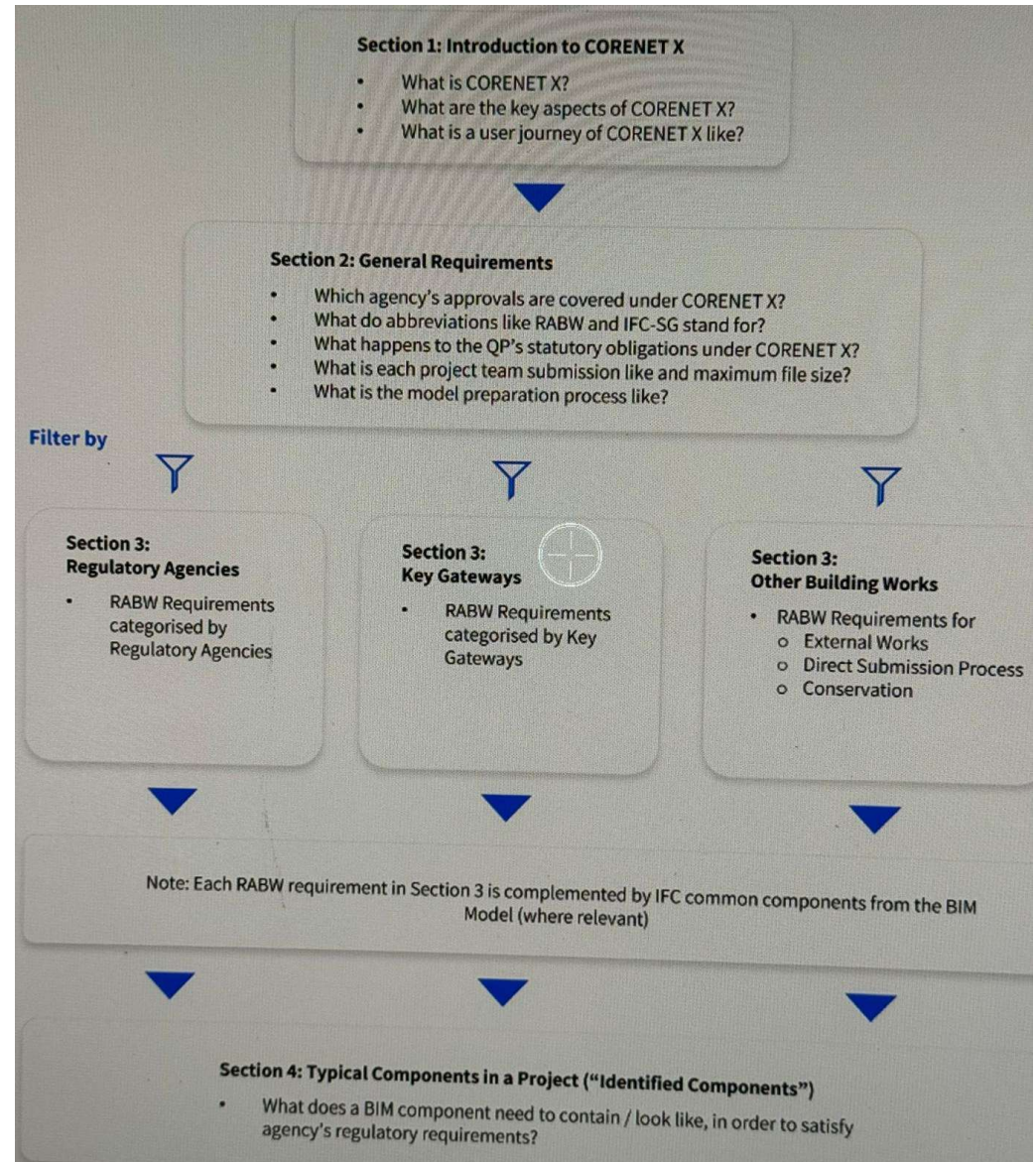
**Next Process:
Chain using Agent
Builder**

BIM 3D Object → 5D QSbar/Spec → 6D IFM

BCA CorenetX

Using IFC-SG

Customised Industry Foundation Class of 3D CAD for Singapore



CoLab drawn by AI

n of Programme 80011 - Imperial College London-BCA E...

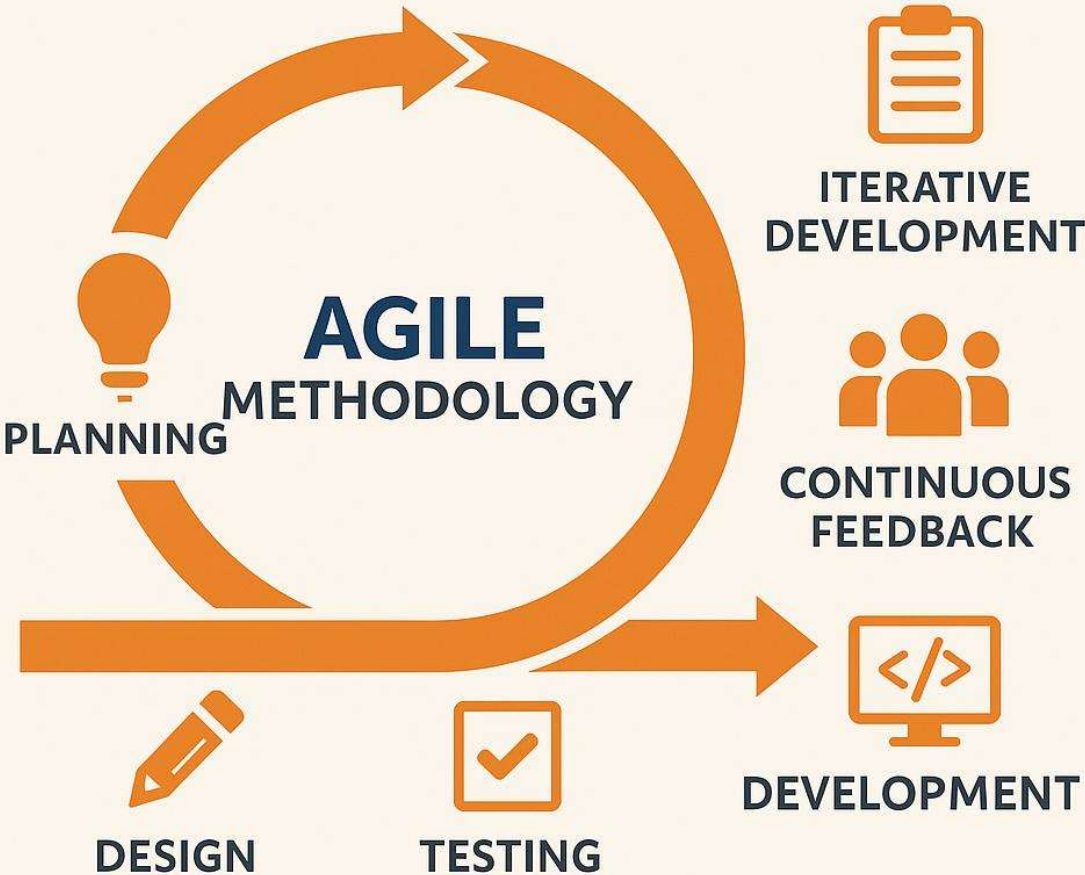
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PROJECT MANAGER
FSISV ICECA MRICS MSc, BSc Bldg Dip Business Admin

CYRIL K. H. SEAH ARCHITECTS

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FAX: (65) 6221-20-59
Mobile: (65) 9643-64-60
E-MAIL: skyey1@yahoo.com



AGILE METHODOLOGY



ATS Advisory Services: A Multi-disciplinary Team

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- 4. Jacky CHIN Kong Hin (Quantity Surveyor) konghinchin@gmail.com
- 5. LIM Wai Sing (Civil & Structural Engineer) limwsgransford@gmail.com
- 6. Augustine YEE Meng Yew (Valuation/Facilities Manager) augustyee@gmail.com
- 7. Daniel CHOO (M&E Engineer) danielchoo@soteria.sg

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ATS Advisory Services
Project Management -
ANG Tok Seng

Multi-disciplinary

Architects/
Interior Designers

Building Surveyors /
Quantity Surveyors

Valuers/
Facilities Managers

Public Relation
Management/
Management Trainer

Engineering Resource
affiliation

Architects -
Cyril K H SEAH Architects

Building Surveyors -
SEAH Kwee Yong

Valuation Surveyors -
Augustine YEE Meng Yew

Public Relation Advisory -
NG Sin Ke

CMP Consultants:
Civil , Structural,
Mechanical and Electrical
ANG Tok Seng (Business
Development)

BIM Managers -
SEAH Kwee Yong

Quantity Surveyors
SEAH Kwee Yong
Jacky CHIN Kong Hin
LEE Fang Cheng

Smart Facilities
Manangers -
Gary ANG Choon Li

Trainer of Design Thinking
-
NG Sin Ke

Soteria Consultants:
M&E
Daniel CHOO

Interior Designers-
Bernard SEAH
Graphic Image Consultant
Joseph SEAH Kwee Hong



Google AI TrailBlazer Demo Slides on IFC CSV

ATS Team



IFC Model

Filter Model

Filter Properties

IDS

Data Analysis

About App

IFC Model Quality Checking Tool





IFC

IFC Model

Filter Model

Filter Properties

IDS

Data Analysis

About App

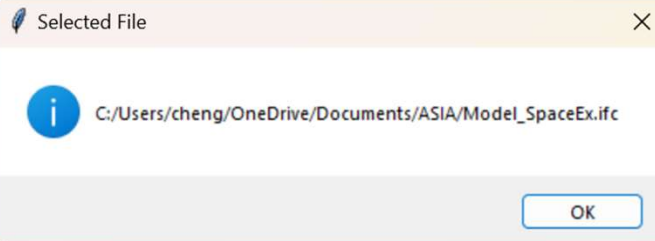
Load IFC

View Model

View Data

Validate Data

Load IFC



Search



ENG
US



5:11 pm
5/4/2025



IFC Model

Filter Model

Filter Properties

IDS

Data Analysis

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Load IFC

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Validate Data

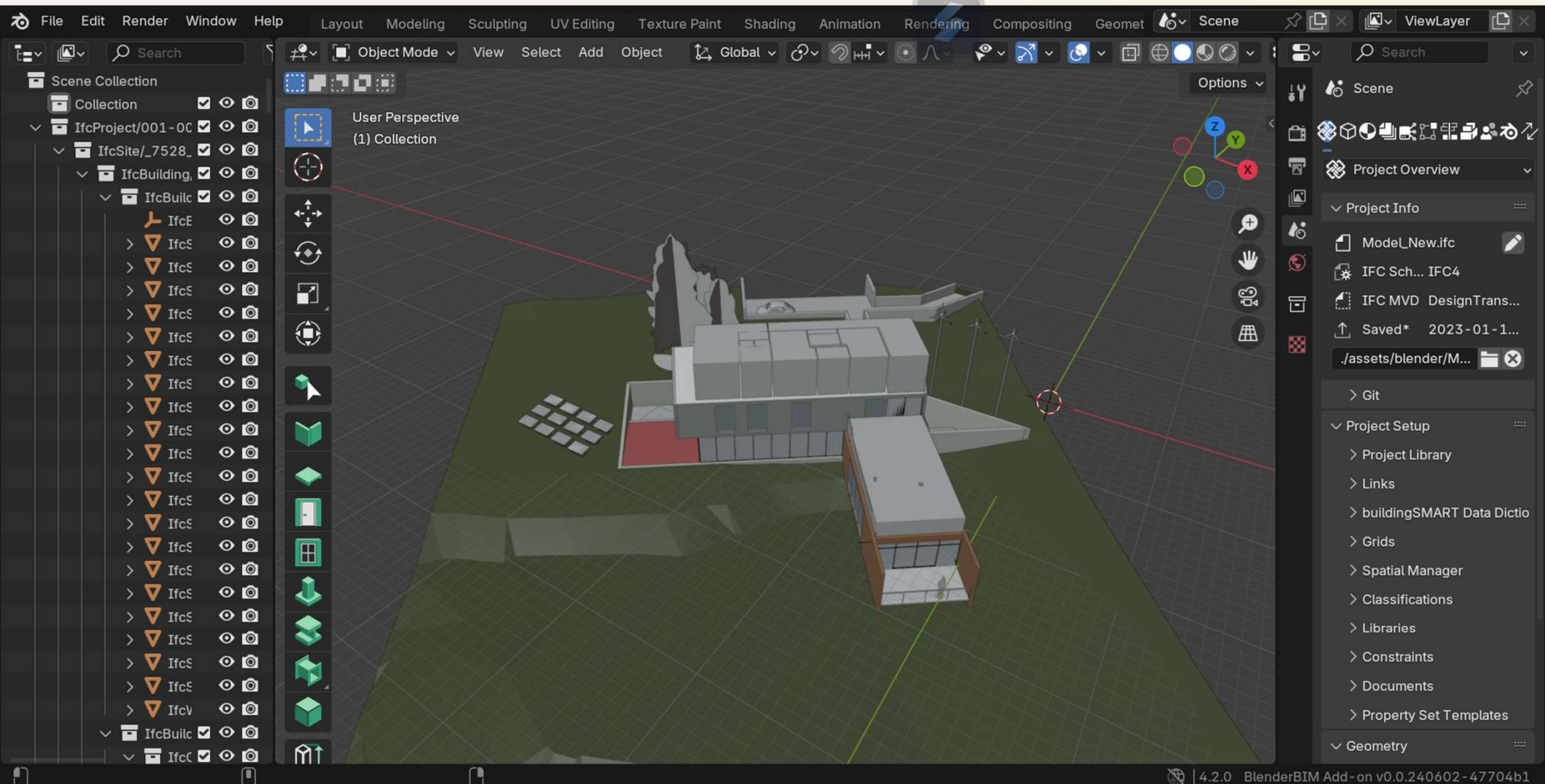
Load IFC

	ExpressId	GlobalId	Class	Predefine	Name	Classifica	Material	Level	Type	Constraint	Pset_Stai	Electric
1	145	3ZGD7y6S	IfcBuilding		Samuel Ma							
2	160	3ZGD7y6	IfcBuildin	Level:8m	Foundatio							
3	166	3ZGD7y6	IfcBuildin	Level:8m	Level 1 Li							
4	170	3ZGD7y6	IfcBuildin	Level:8m	Level 1							
5	176	3ZGD7y6	IfcBuildin	Level:8m	Ceiling							
6	182	3ZGD7y6	IfcBuildin	Level:8m	Level 2							
7	188	3ZGD7y6	IfcBuildin	Level:8m	Roof Line							
8	7528	3ZGD7y6S	IfcSite		Topograph							
9	7636	3ch3OBg	IfcSpace	SPACE	206				Rooms : M			
10	7825	3ch3OBg	IfcSpace	SPACE	207				Rooms : M			
11	7970	3ch3OBg	IfcSpace	SPACE	204				Rooms :			
12	8098	3ch3OBg	IfcSpace	SPACE	203				Rooms : B			
13	8226	3ch3OBg	IfcSpace	SPACE	205				Rooms : B			
14	8369	3ch3OBg	IfcSpace	SPACE	202				Rooms :			
15	8514	3ch3OBg	IfcSpace	SPACE	201				Rooms : En			
16	8698	3ch3OBg	IfcSpace	SPACE	101				Rooms : K			
17	8828	3ch3OBg	IfcSpace	SPACE	104				Rooms : L			
18	8956	3ch3OBg	IfcSpace	SPACE	103				Rooms : B			
19	9107	3ch3OBg	IfcSpace	SPACE	105				Rooms : H			
20	9235	3ch3OBg	IfcSpace	SPACE	106				Rooms : L			
21	9363	2pfAHb2E	IfcSpace	SPACE	102				Rooms : M			
22	9491	2wm6PO6	IfcSpace	SPACE	208				Rooms : L			



Q Search

ENG
US5:11 pm
5/4/2025





Select Entity Filter psets Filter qtos Filter psets rules Filter qtos rules View filtered entities Gemini AI

Select Entity

- IFC Model
- Filter Model
- Filter Properties
- IDS
- Data Analysis
- About App

Management Dashboard

Filter Elements By Property/Quantity Sets

Foundation	IfcSlab
Level 1 Living Rm.	IfcWall
Level 1	IfcOpeningElement
Ceiling	IfcFlowTerminal
Level 2	IfcRoof
Roof Line	IfcColumn
	IfcDoor
	IfcCurtainWall
	IfcPlate
	IfcMember

Level 1 is selected IfcWall is selected



Search



ENG
US



5:13 pm
5/4/2025



IFC Model

Filter Model

Filter Properties

IDS

Data Analysis

About App

Select Entity

Filter psets

Filter qtos

Filter psets rules

Filter qtos rules

View filtered entities

Gemini AI

Select Entity

Selected floor : Level 1

Selected entity : IfcWall

AI csv ouput



Outputdf.csv file created successfully. You can now run Gemini AI

OK



Search



ENG
US



10:49 am
7/4/2025

Gemini AI Prompt

```
- Classification
- Level
- Qto_WallBaseQuantities.id
- Qto_WallBaseQuantities.GrossSideArea
- Qto_WallBaseQuantities.GrossFootprintArea
- Qto_WallBaseQuantities.NetVolume
- Qto_WallBaseQuantities.NetSideArea
- Qto_WallBaseQuantities.Height
- Qto_WallBaseQuantities.Width
- Qto_WallBaseQuantities.Length
- Qto_WallBaseQuantities.GrossVolume
```

```
-----
```

```
Enter your query about the CSV data (e.g., 'What is the largest GrossSideArea?', 'Show walls on Level 1'):
>
```

Show specific columns – save to output.csv

```
- Classification
- Level
- Qto_WallBaseQuantities.id
- Qto_WallBaseQuantities.GrossSideArea
- Qto_WallBaseQuantities.GrossFootprintArea
- Qto_WallBaseQuantities.NetVolume
- Qto_WallBaseQuantities.NetSideArea
- Qto_WallBaseQuantities.Height
- Qto_WallBaseQuantities.Width
- Qto_WallBaseQuantities.Length
- Qto_WallBaseQuantities.GrossVolume
```

Enter your query about the CSV data (e.g., 'What is the largest GrossSideArea?', 'Show walls on Level 1'):

> Show all rows with column headers "Level", "Height" and "Width"

[illegible]

Thank you

Proof of Concept (PoC) and **Minimum Viable Product (MVP)** are two important concepts in product development:

PoC: Aims to demonstrate the technical feasibility of an idea. It is used to validate whether a concept can be developed into a functional product.

MVP: Focuses on creating a functional version of a product with core features to test market demand and gather user feedback. It helps in understanding if there is a viable market for the product.

In summary, PoC is about proving an idea can work technically, while MVP is about validating market interest and usability with real users.